



Hochschule Aalen

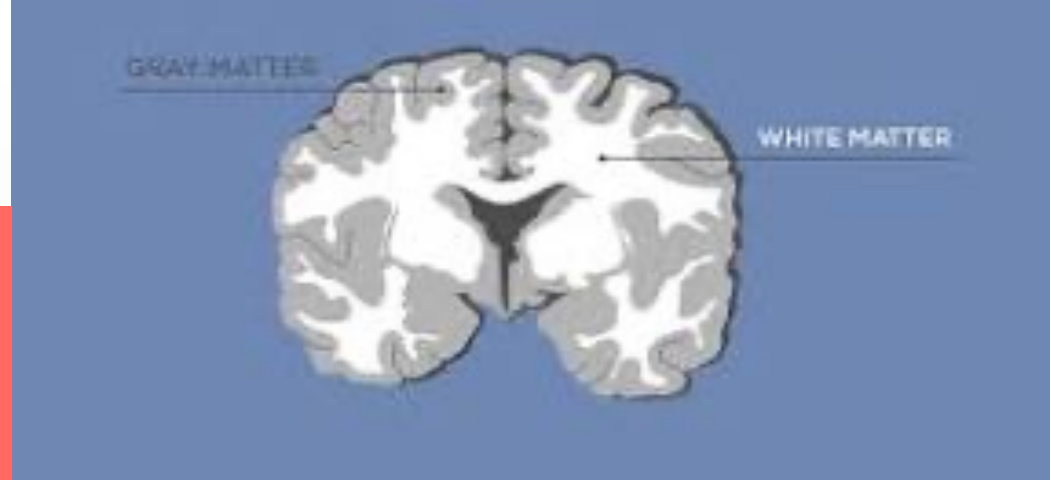
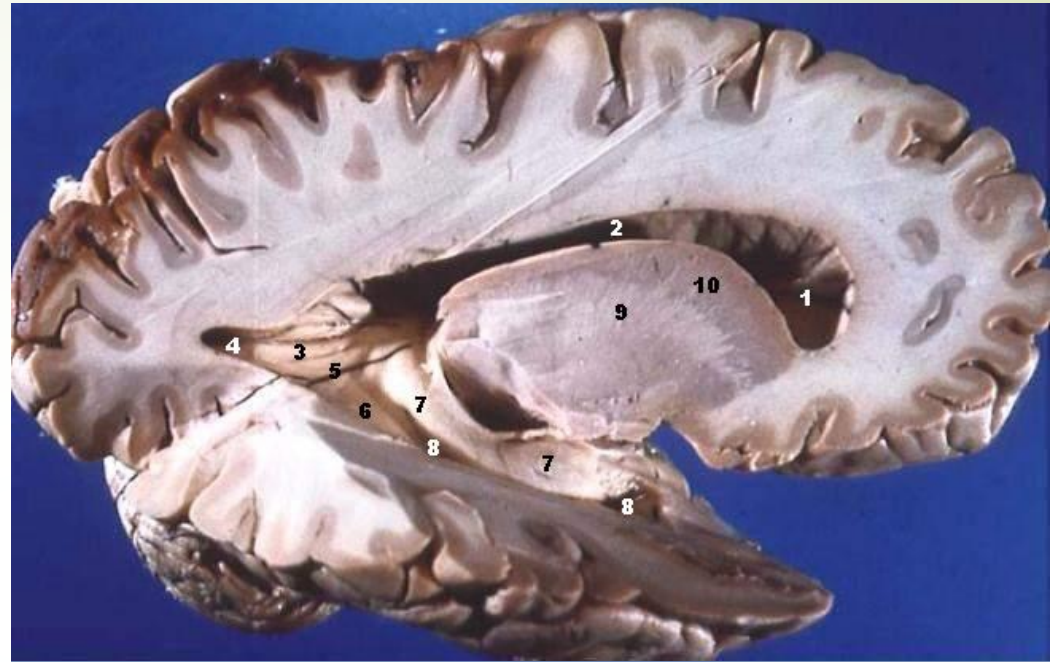
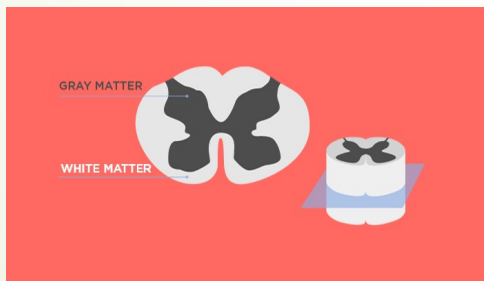
Neural Cells and Glia



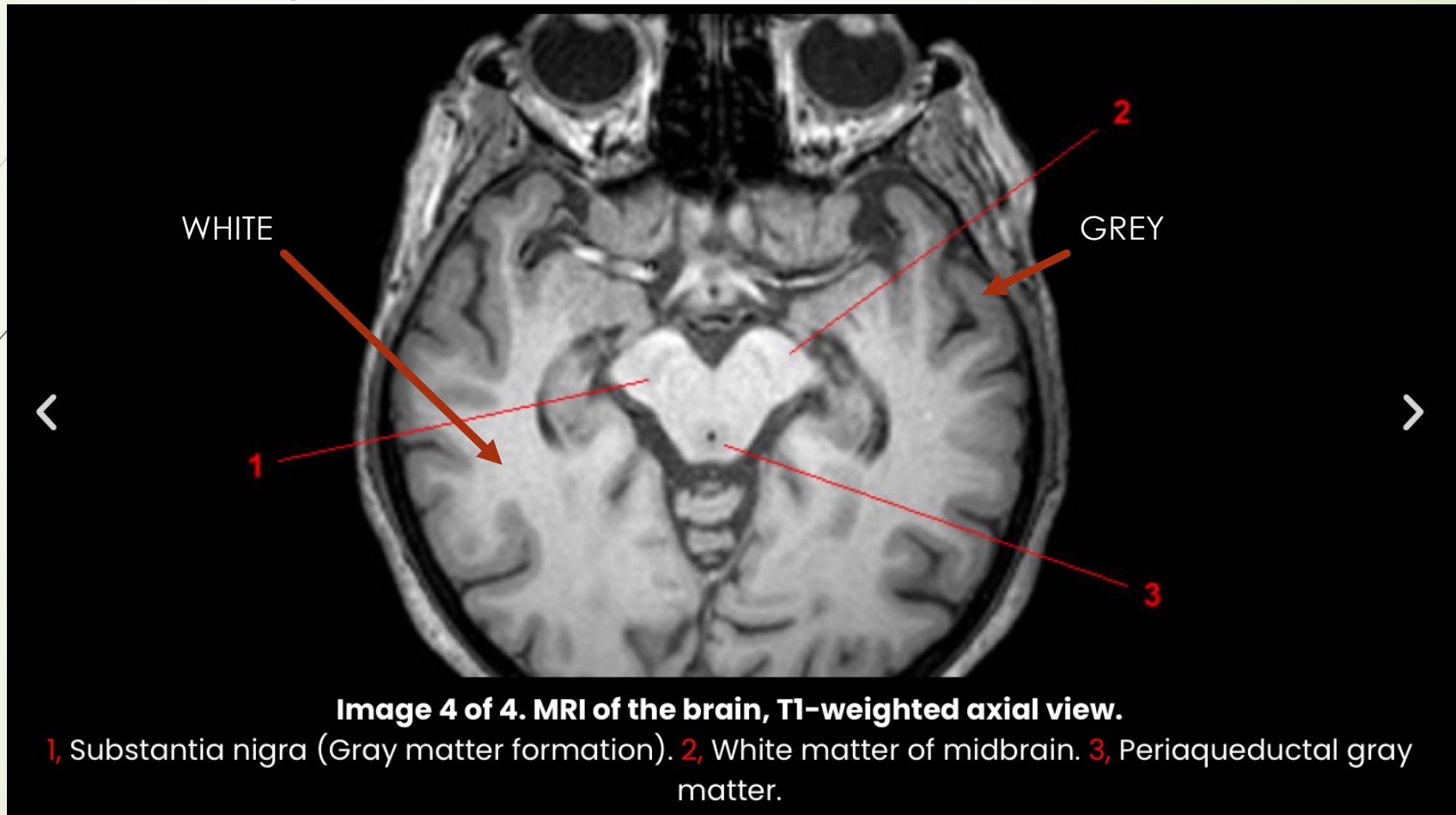
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White Matter vs Grey Matter

- Grey matter is made up of neuronal cell bodies, while white matter consists mostly of myelinated axons. The myelin is what gives the white color. In the brain, white matter is found closer to the center of the brain, whereas the outer cortex is mainly grey matter. In the brain the grey is on the outside. Spinal cord its on the inside.



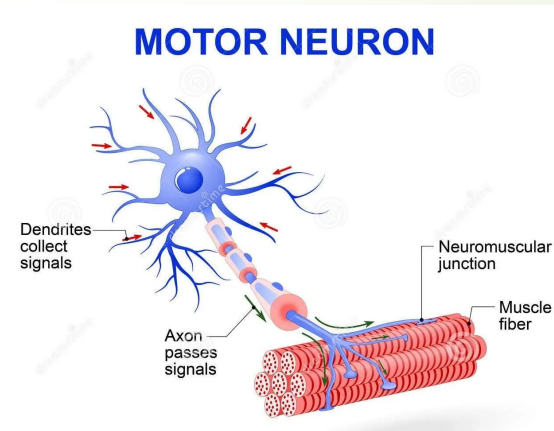
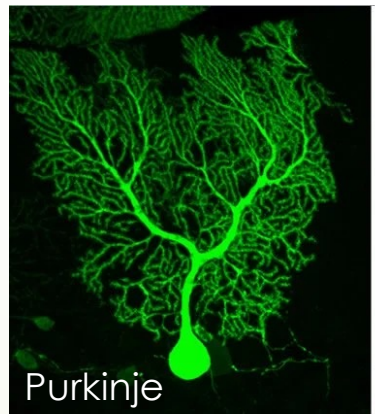
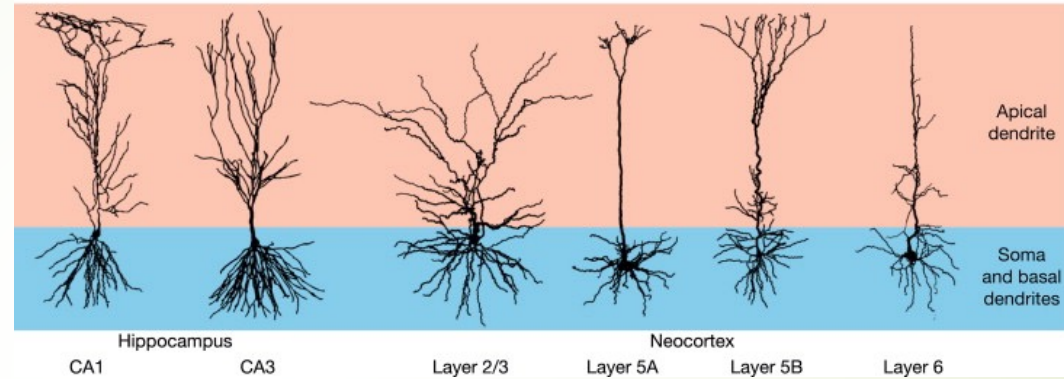
MRI (T1-weighted) allows imaging of white vs grey matter



“Typical” neurons

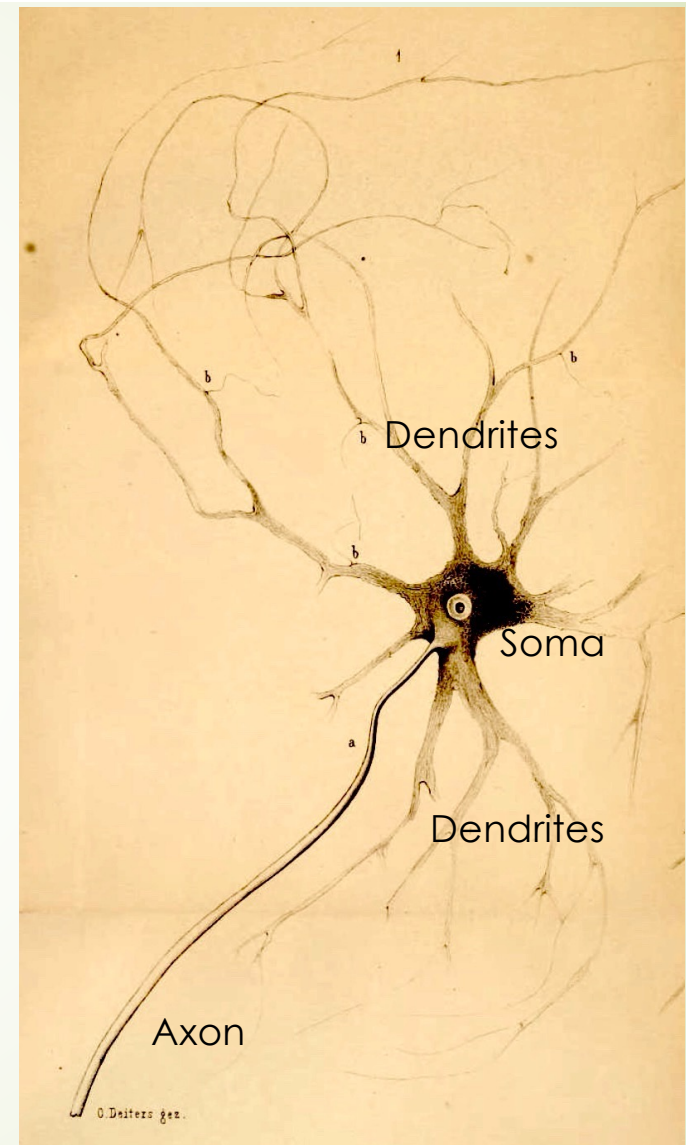
- Nerve cells with the longest, largest diameter axons are often illustrated as “typical” neurons. These include -- pyramidal cells of motor cortex and motor neurons of spinal cord. Cerebellar Purkinje cells comprise another type of “typical” nerve cell, also large but with a huge dendritic tree rather than an especially long axon.

Pyramidal cell types Pyramidal cell variations



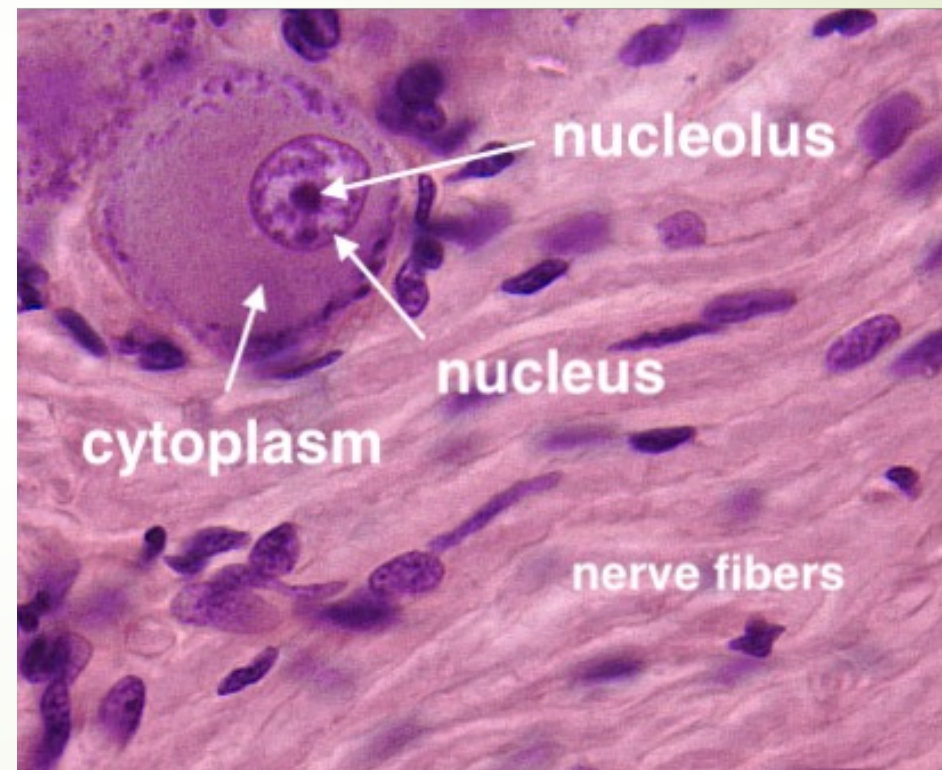
Typical Neurons

- Cell Body (soma)
- Axon (only one – or none, like amacrine cells in the retina)-this conveys the cell signal going out.
 - Myelinated
 - Unmyelinated
- Dendrites (many). These are the inputs to the cell.

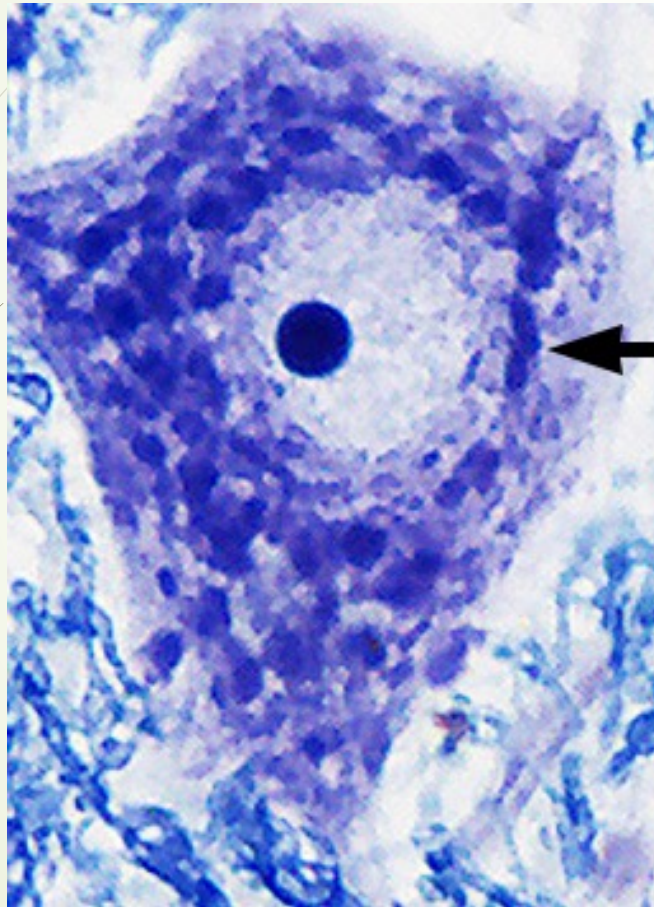


The Cell Body (Soma)

- The **cell body** of a nerve cell (aka *perikaryon* or *soma*) is a cell nucleus surrounded by cytoplasm. Both axons and dendrites extend from this part of the cell.
- Distinguishing features, compared to other cells, include:
 - **Nuclei** of nerve cells are large, round and euchromatic with a single prominent nucleolus.
 - **Cytoplasm** is commonly abundant and basophilic, (i.e. stains dark purple or blue) containing characteristic basophilic masses of rough endoplasmic reticulum that when seen by light microscopy are called **Nissl bodies** (after [Franz Nissl](#), b. 1860).



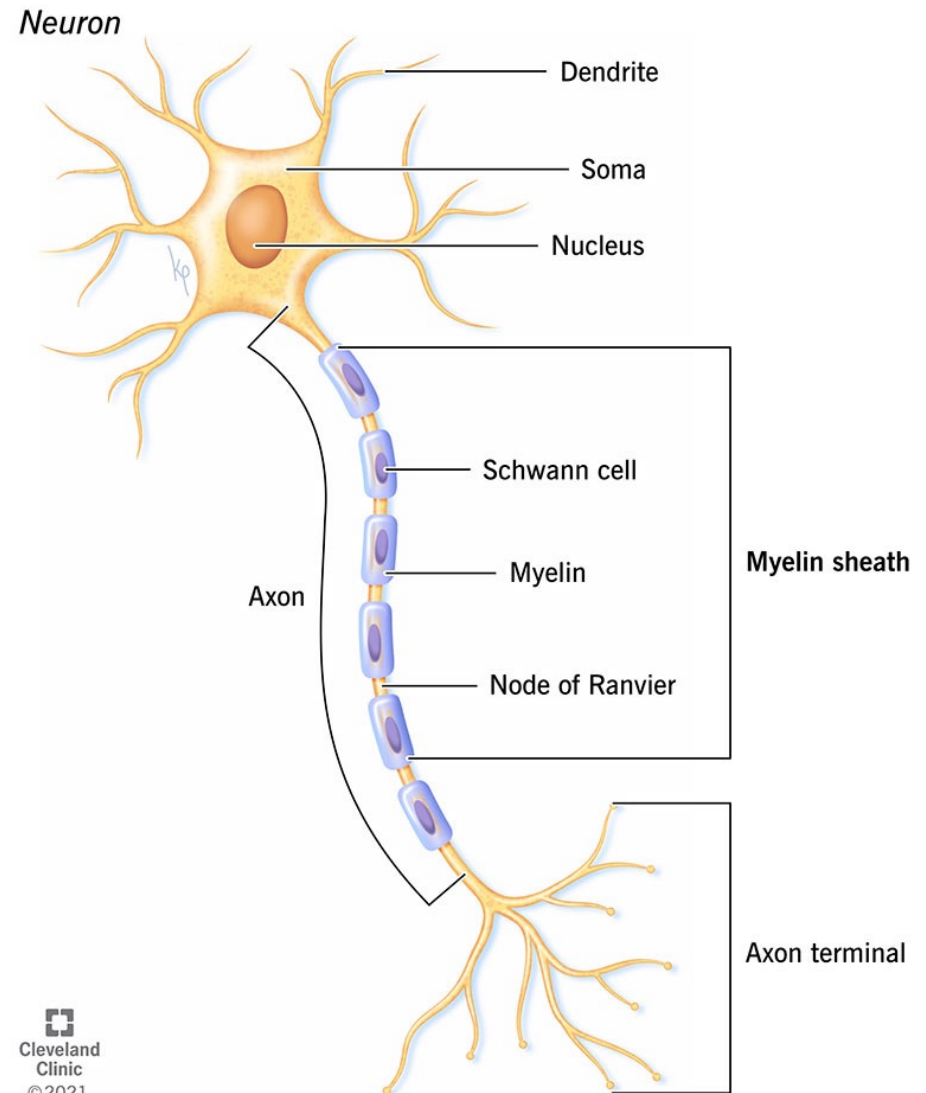
Nissl Bodies (aka Nissl substance)



- Clumps of rough endoplasmic reticulum in the cytoplasm of a neuron, as seen by light microscopy.

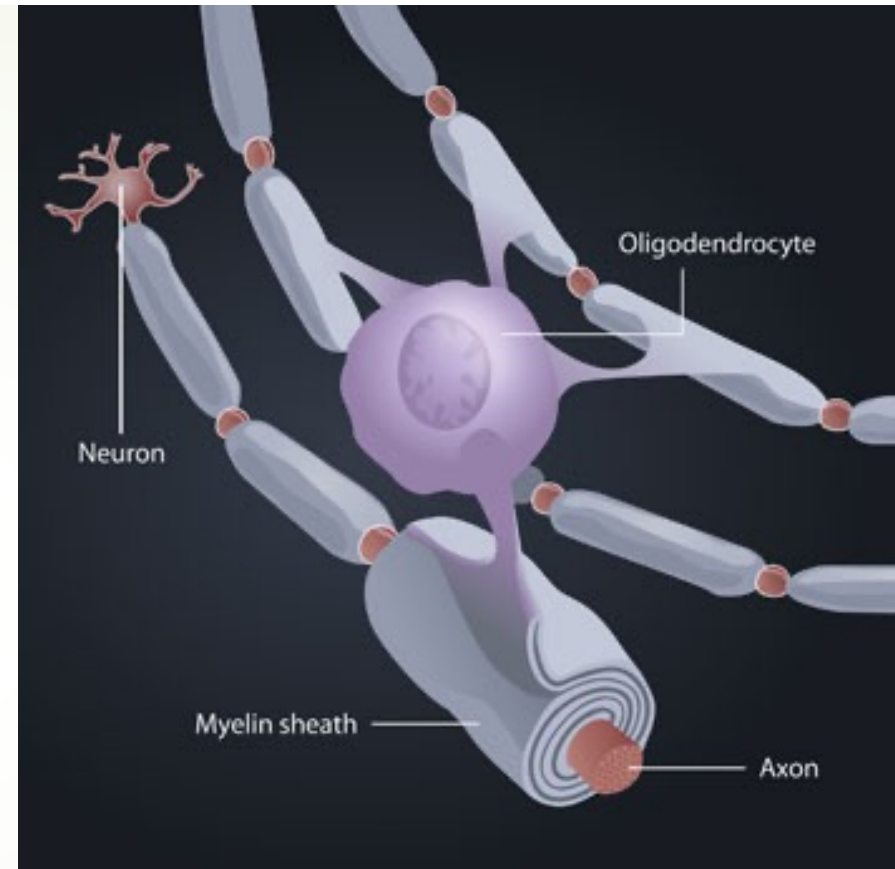
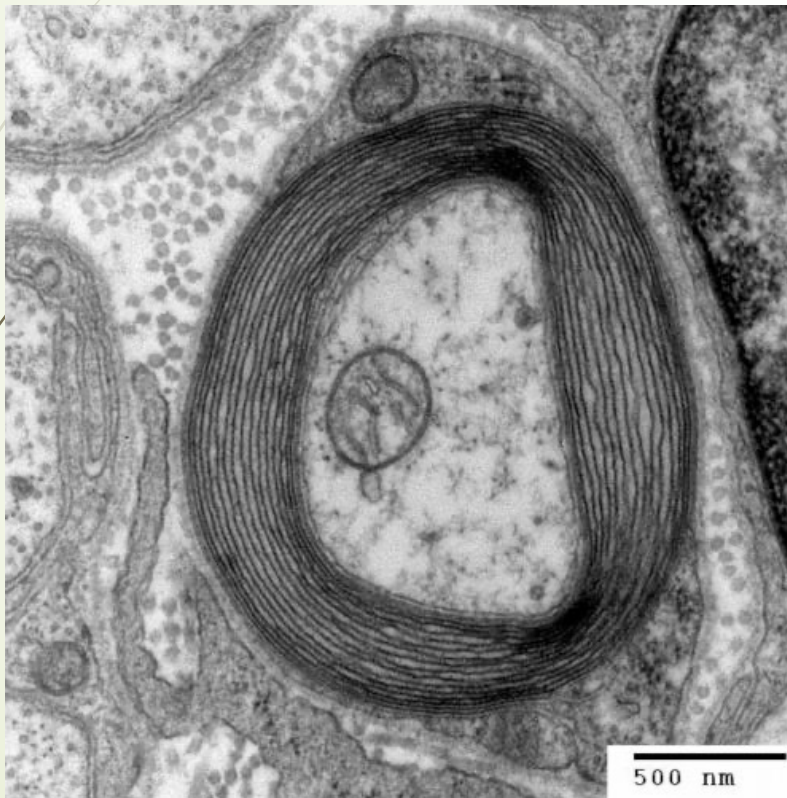
MYELINATED AXON

- Myelinated nerve axons are wrapped in spiralling layers of cell processes that contain an insulating material called myelin.
- In the CNS, the cells that produce myelin are glial cells called oligodendrocytes. In the peripheral nervous system, these axons are wrapped in myelin sheaths provided by Schwann cells. Peripheral nerves can regenerate. CNS cells cannot.
- Between myelin wrapped areas are short, unmyelinated segments called Nodes of Ranvier.
- Speed of transmission is much greater in myelinated axons because membrane depolarization, used to propagate the cell signal, only occurs at the nodes. In unmyelinated nerves the entire length of the axonal cell membrane must transmit the depolarization.



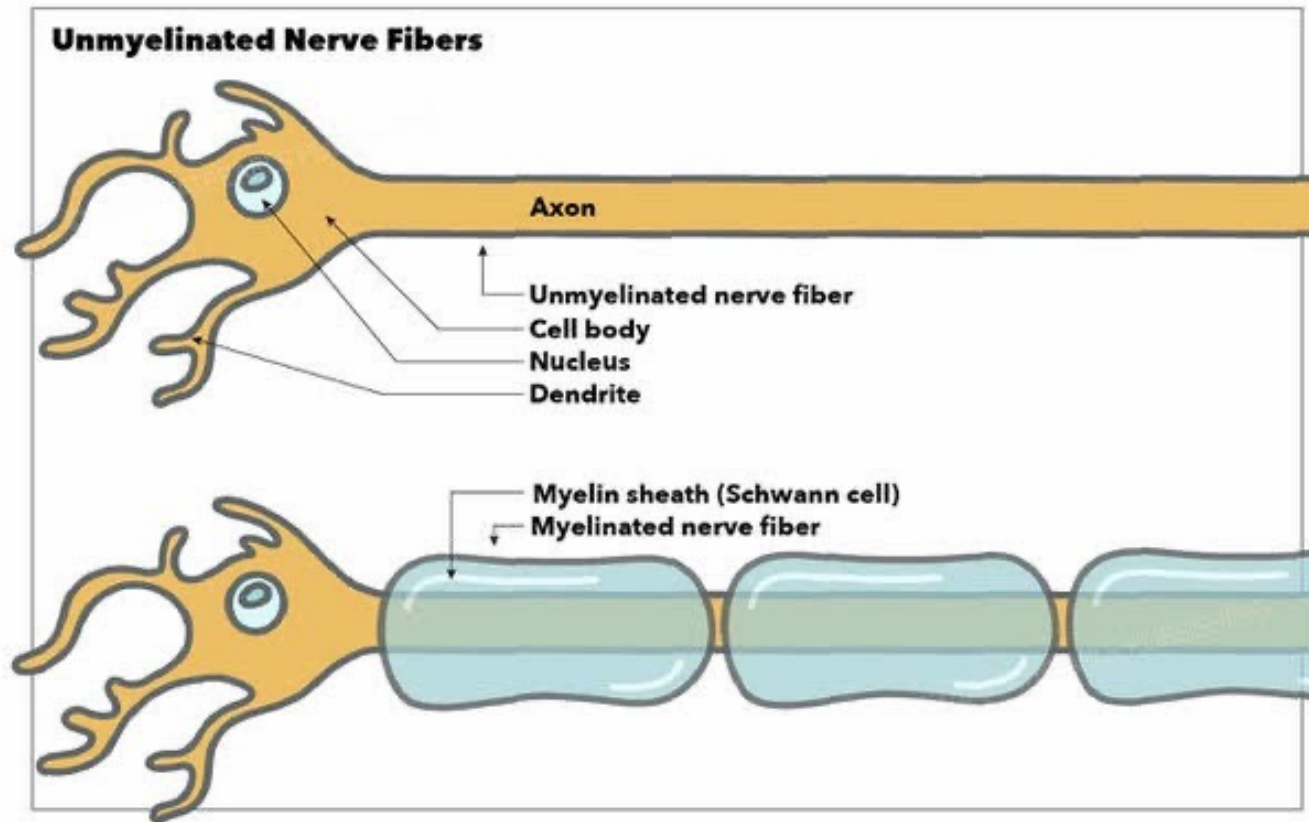
Myelination

Electron micrograph of successive layers of myelin wrapping provided by an oligodendrocyte



Relationship between oligodendrocyte and processes that wrap axons in insulating myelin sheaths. Oligos, as shown can myelinate several cells. In the peripheral nervous system, Schwann cells are devoted to wrapping only a single cell.

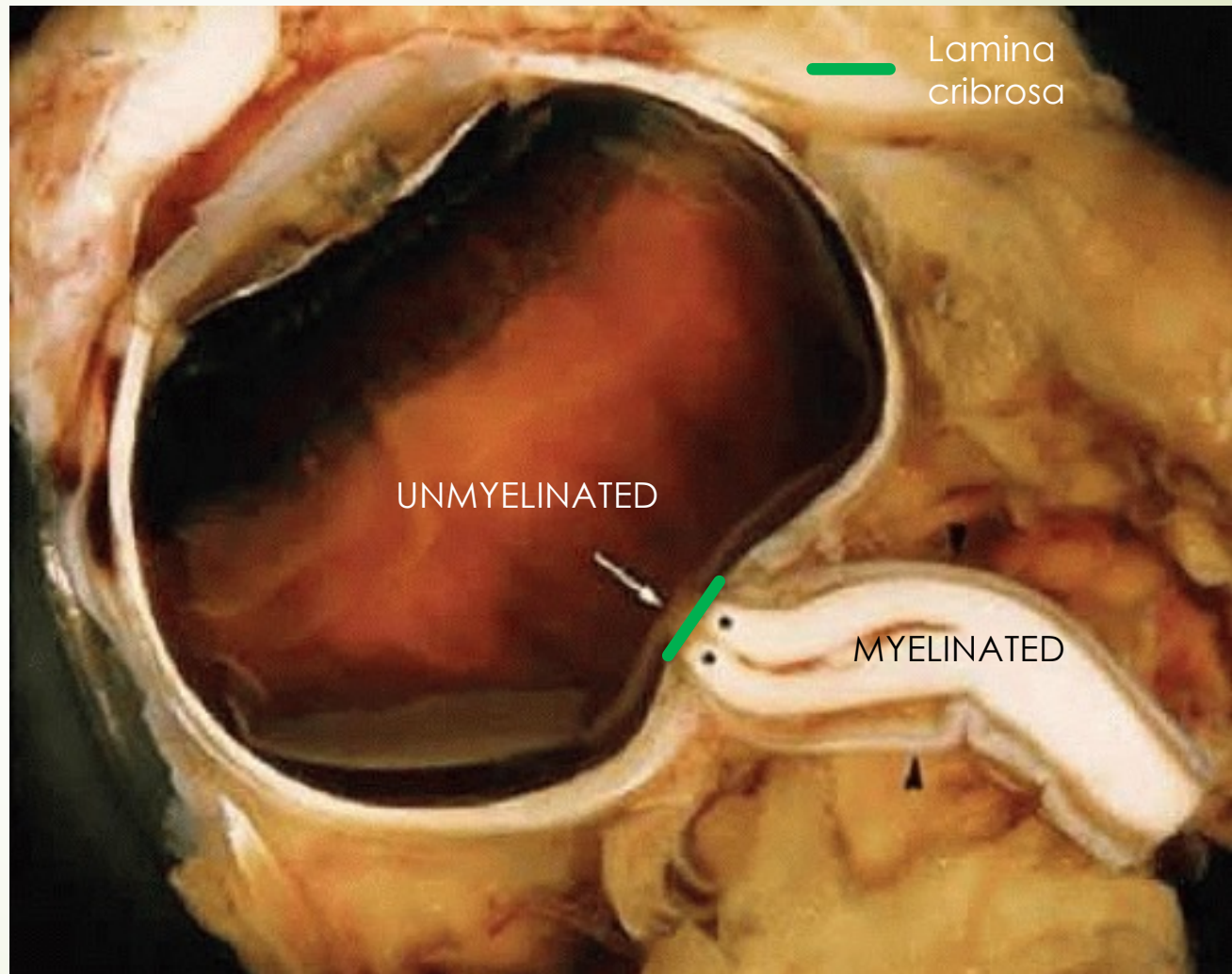
UNMYELINATED AXON



- They conduct low-velocity impulses, and represent the majority of peripheral sensory and autonomic fibers.

An Axon that is BOTH MYELINATED AND UNMYELINATED

- ▶ The axons of Retinal Ganglion cells begin unmyelinated. Myelin is opaque and with axons running on the inner surface of the retina, they would block light from reaching photoreceptors. But as soon as axons leave the retina to join the optic nerve (at the lamina cribrosa), they become myelinated, rapidly enhancing transmission speed.
- ▶ This individual wrapping of each axon doubles the diameter of the optic nerve from 1.5 mm at the optic nerve head, to 3 mm as the nerve exists the sclera.



Myelinated Retinal Nerve Fiber Layer

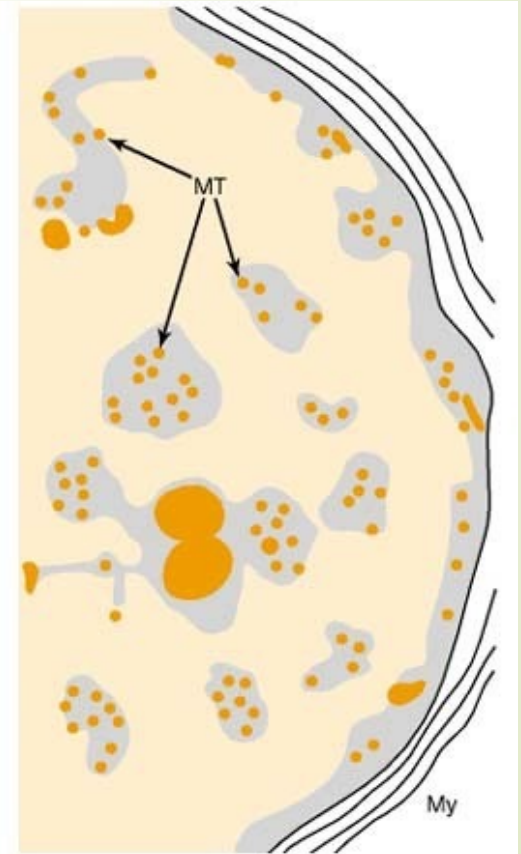
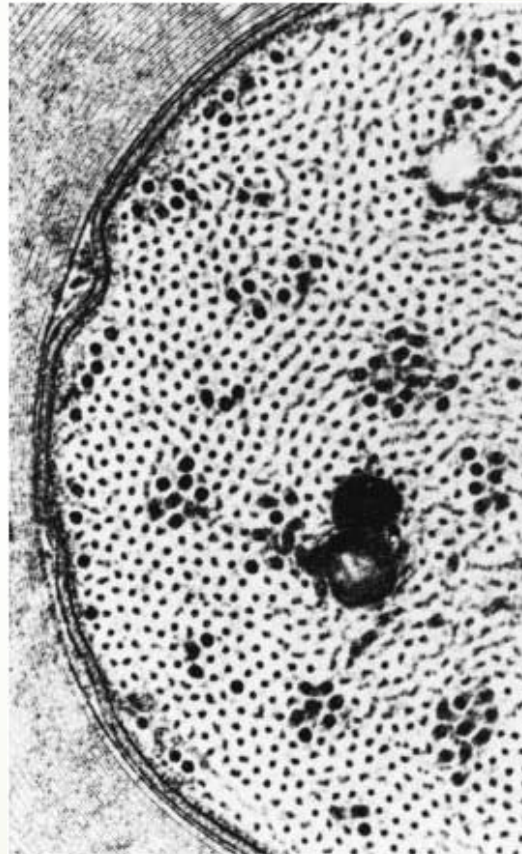


- This is a congenital condition. Whether it is genetic in origin is uncertain. Because the myelin is opaque, you can map an absolute scotoma that has the same shape as the myelin distribution.
- Generally, the macular area is not involved. And the involved area remains unchanged.
- Rare cases of progression have been reported. If the myelinated axons die (e.g. glaucoma), they lose their myelin and so the area can be seen to be regressing.

CYTOPLASMIC FILAMENTS

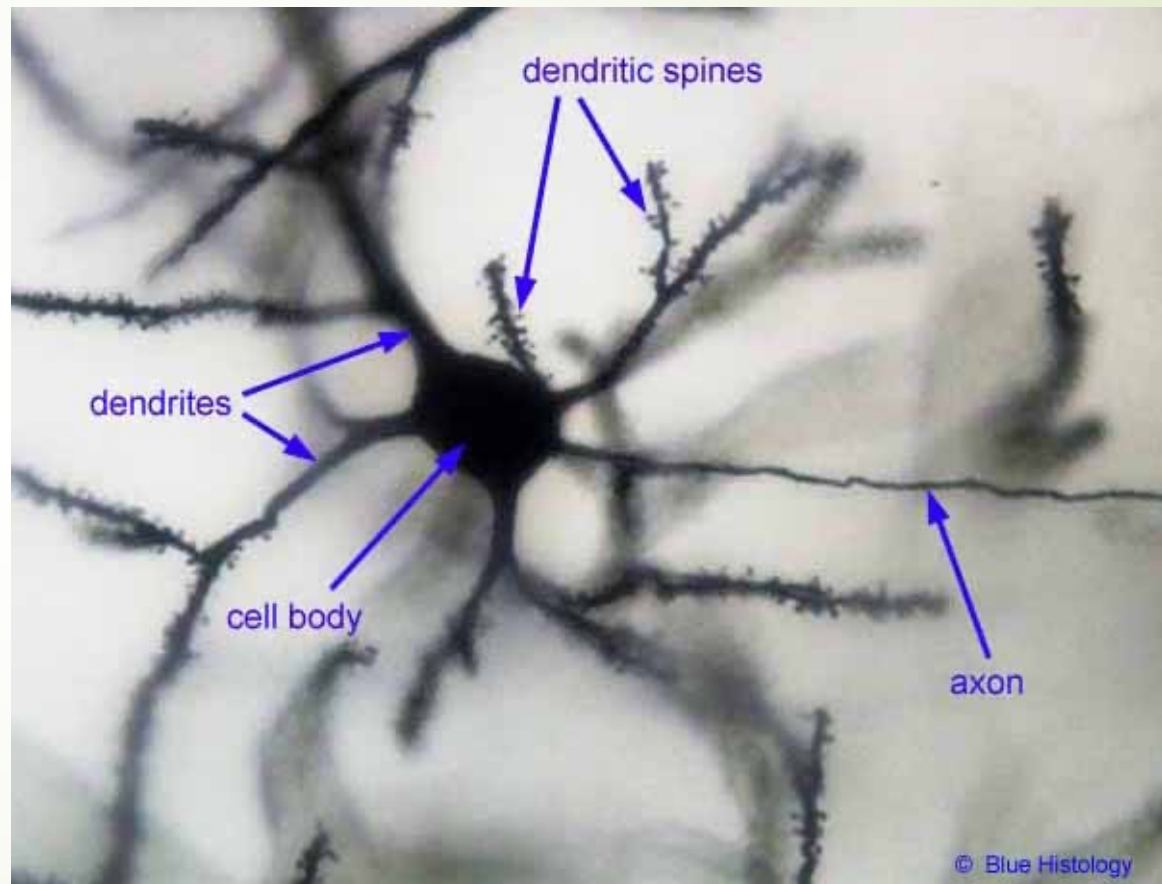
- Like other eukaryotic cells, neurons have a cytoskeleton that consists of three main polymers:
 - Microtubules
 - Intermediate filaments (aka neurofilaments)
 - actin microfilaments
- Neurofilaments (*clear areas*). The microtubules (MT) tend to be found in bundles and are more irregularly spaced. They are surrounded by a fuzzy material thought to be enriched in actin microfilaments.

Cross-section of an axon



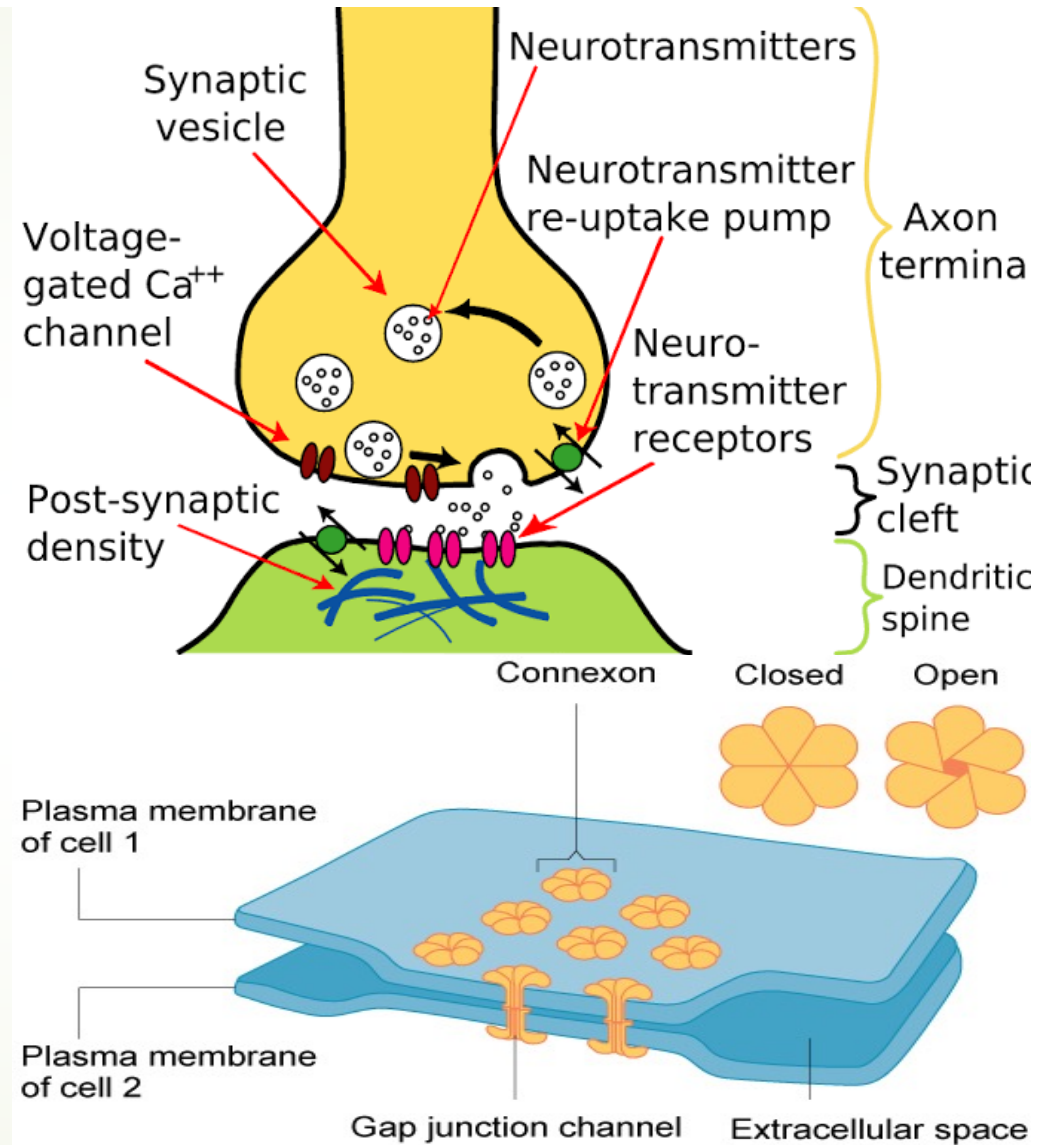
DENDRITES

- Dendrites are the parts of the neuron that receive inputs from axons of other cells via synapses.
- Dendrites may extend up to a millimeter or two away from the cell body, seldom more.
- Dendrites may be myelinated or unmyelinated. Myelinated ones are more likely to occur in the peripheral nerves that must conduct impulses for long distances.



SYNAPSES

- Chemical synapses occur at the end of axons. They produce, release and reabsorb various neurotransmitter molecules that bind to a receptor on the cell to which the axon is joined. This transmitter may stimulate or inhibit the cell to which the synapse is joined.
- Electrical synapses convey stimuli by passing electrical current through low resistance gap junctions

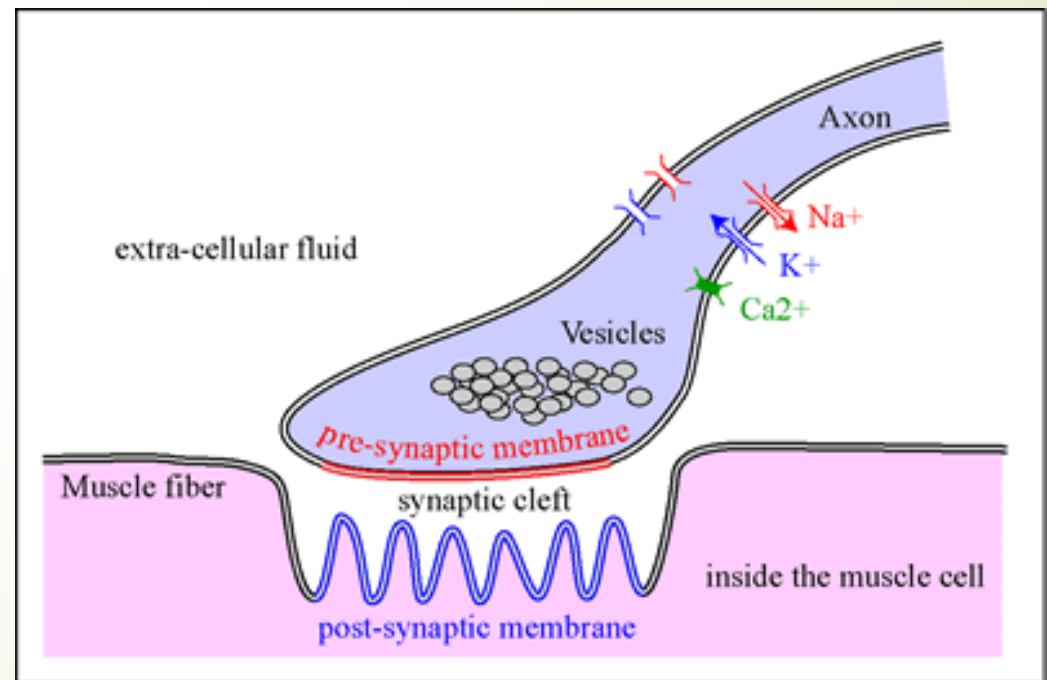
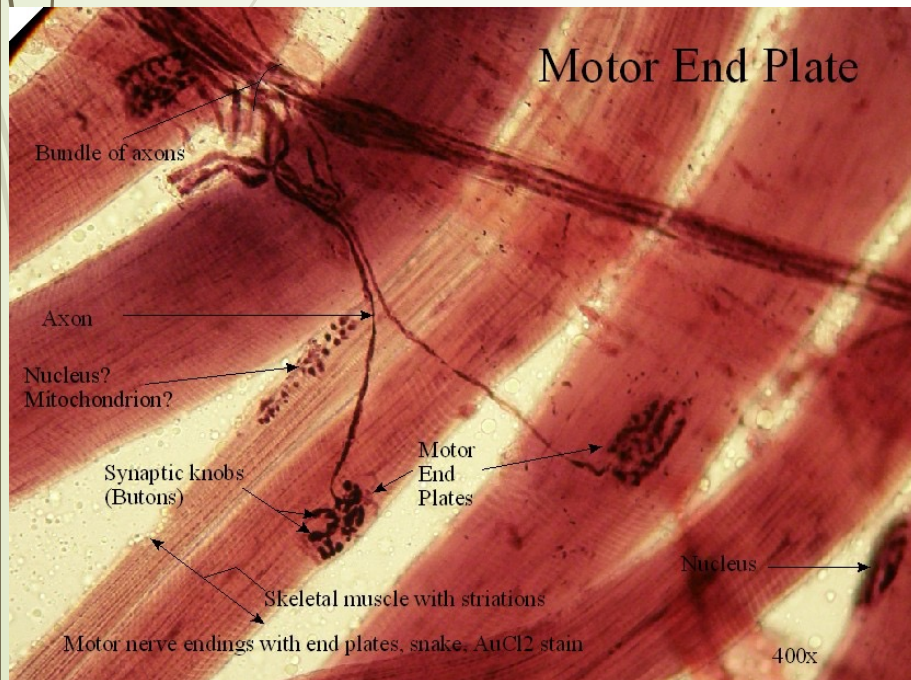


AXOPLASMIC FLOW

- The neuronal cell has to transport proteins, lipids, other macromolecules and even organelles such as mitochondria, from the cell body to the synapses (**anterograde** direction). This is because all of the protein making machinery (e.g. ribosomes, endoplasmic reticulum) is in the cell soma.
- There is a slow transport system moving at a few millimeters a day, and a fast system, which is about 100 times faster.
- There is also **retrograde flow**, in which unused or recycled constituents are returned to the cell body.

MOTOR END PLATES (aka Neuromuscular junctions)

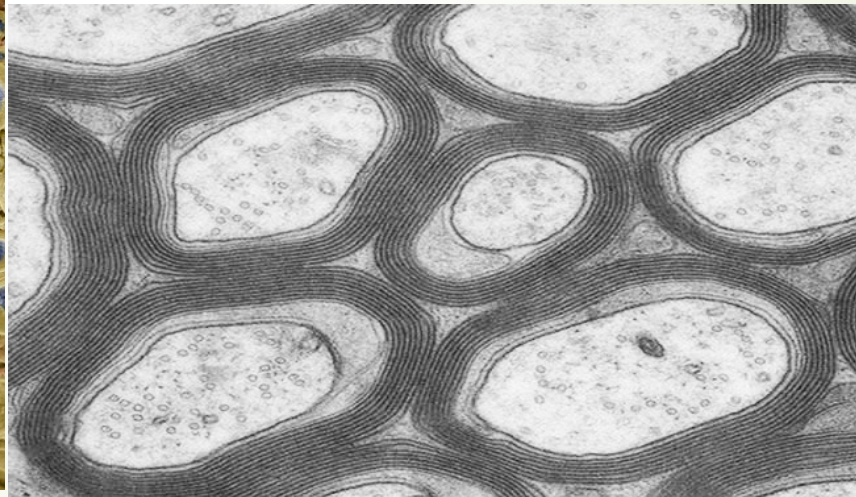
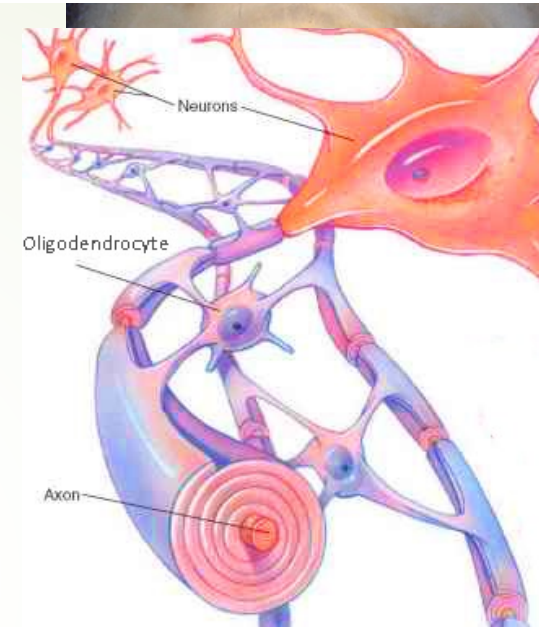
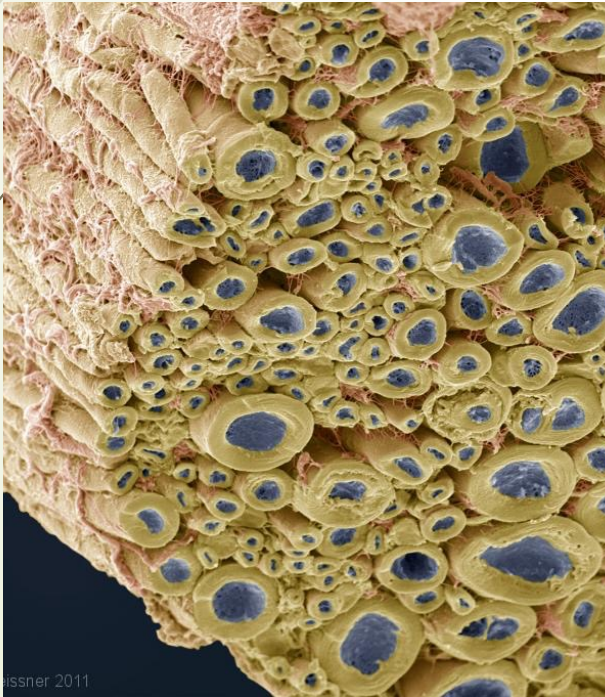
- The neuromuscular junction is a synaptic connection between the terminal end of a motor nerve and a muscle (skeletal/ smooth/ cardiac). All of the muscle cells served by a given neuron are considered a motor unit. The motor end plate is the site for the transmission of an **action potential** from nerve to the muscle. Propagation of the action potential along the cell membrane of the muscle cell, leads to contraction.



Neuroglia

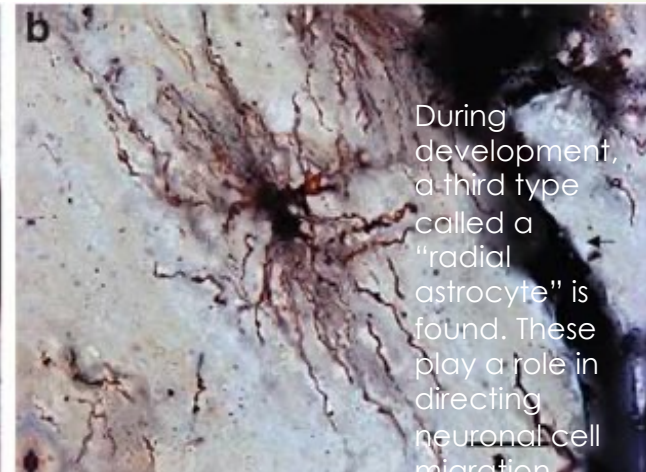
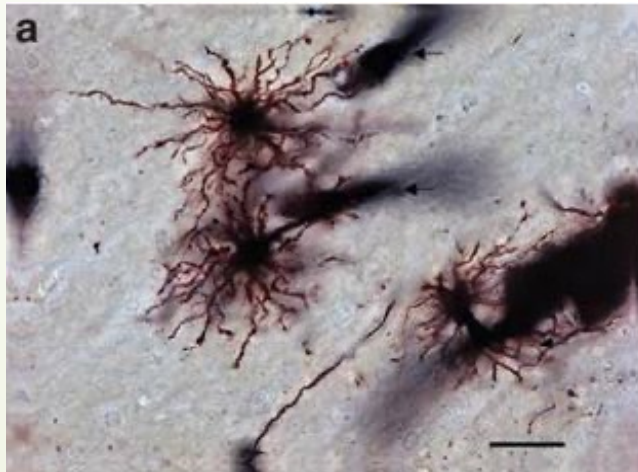
- ▶ **Oligodendroglia** – these cells make myelin in the CNS (recall that this is the job of Schwann cells in the peripheral nervous system).
- ▶ **Astrocytes** – the main regulatory and supportive glial cells, also provide tissue support, Commonly absorb nutrients from blood vessels and convey to neurons.
- ▶ **Microglia** – the reticuloendothelial cells of the CNS. Roles similar to macrophages, including inflammatory modulation, phagocytosis, antigen recognition and processing.

Oligodendrocytes form Myelin in the CNS



Astrocytes and Their Sub-Types

- **Protoplasmic (aka, Type 1, with a and b sub-types)** – found mostly in grey matter, show a more spherical cell body, with tortuous varicose and thorny processes radiating in all directions. (picture a – notice the overall spherical profile). These all exhibit glial fibrillary acid protein (GFAP). Type 1A astrocytes express GFAP but not neural cell adhesion molecule (NCAM). Type 1B astrocytes, however, express BOTH GFAP and NCAM and are the dominant astrocyte in the optic nerve head.
- **Fibrous (aka, Type 2)** – found primarily in white matter. These cells usually exhibit an oblong cell body, with relatively long and unramified varicose and thorny processes extended mainly in two opposite directions. These cells often give rise to “vascular feet” that make connections to the outsides of vascular walls. (picture b, notice the overall elliptical or axial profile)

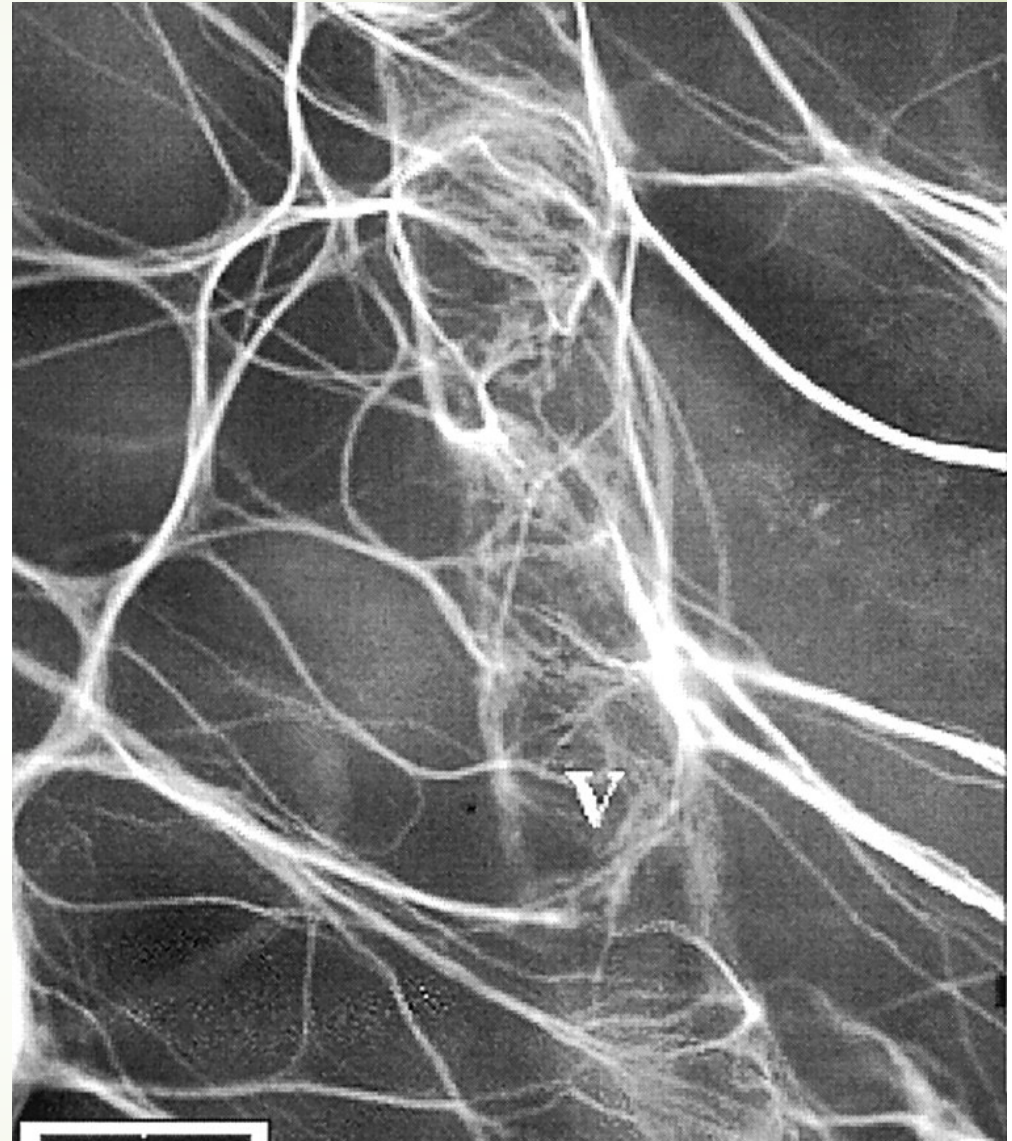


During development, a third type called a “radial astrocyte” is found. These play a role in directing neuronal cell migration.

Astrocytes

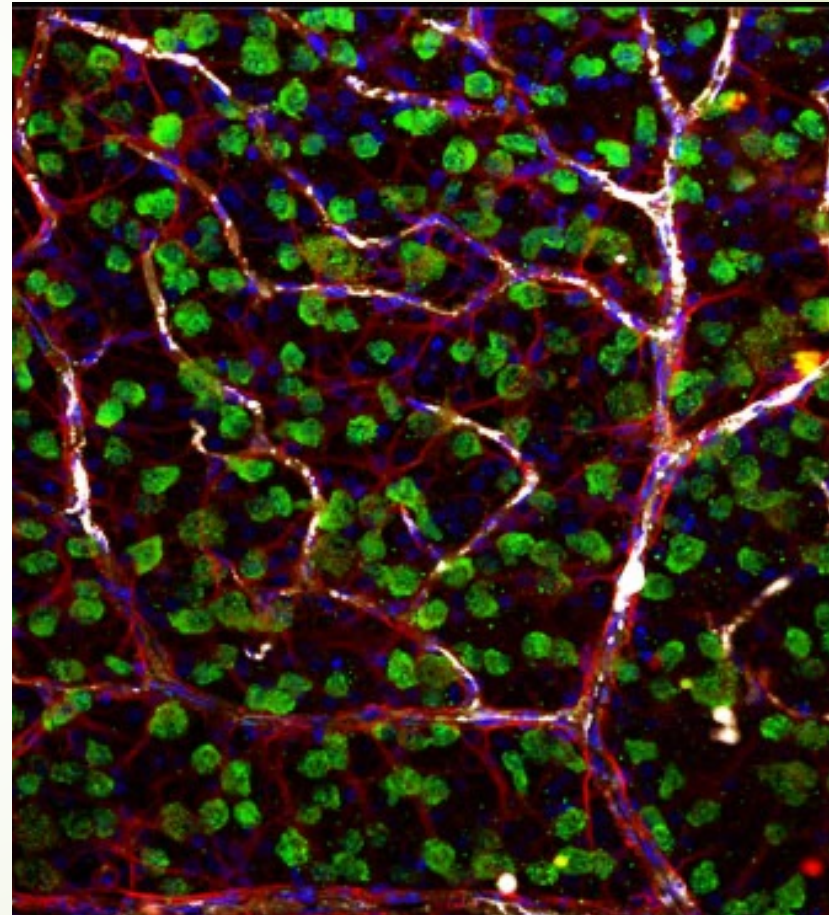
- ▶ Astrocytes communicate between neurons, blood vessels, and other types of glial cells. They form a coordinated syncytium by communicating through numerous gap junctions.

Astrocytes give off numerous fine processes extending to the surface of a blood vessel (V)

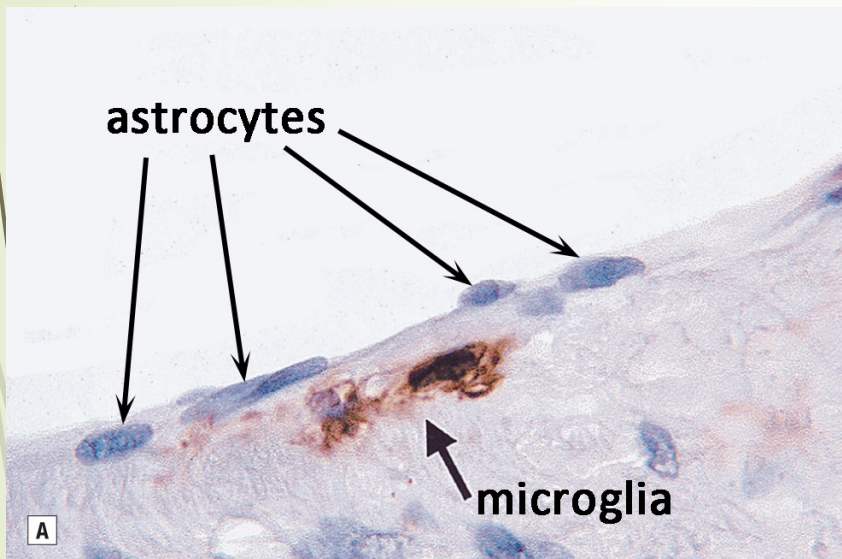


Relationship Between Retinal Ganglion Cells, Astrocytes and Blood Vessels

Note the anatomical relationships between ganglion cells (green), thin astrocyte processes (red) and blood vessels. The traditional notion is that these astrocyte processes surround blood vessels, pick up nutrients and then deliver them to neuronal cells. They also communicate with each other and other glia via gap junctions.



Microglia



Inactive, quiescent astrocytes and microglia at the surface of the optic nerve head

Microglia are the reticuloendothelial system representative in the nervous system, including the retina. As the resident macrophages of the nervous system, they act as the first and main form of active immune defense CNS, including the retina

Resting microglia cannot mount an immune response.

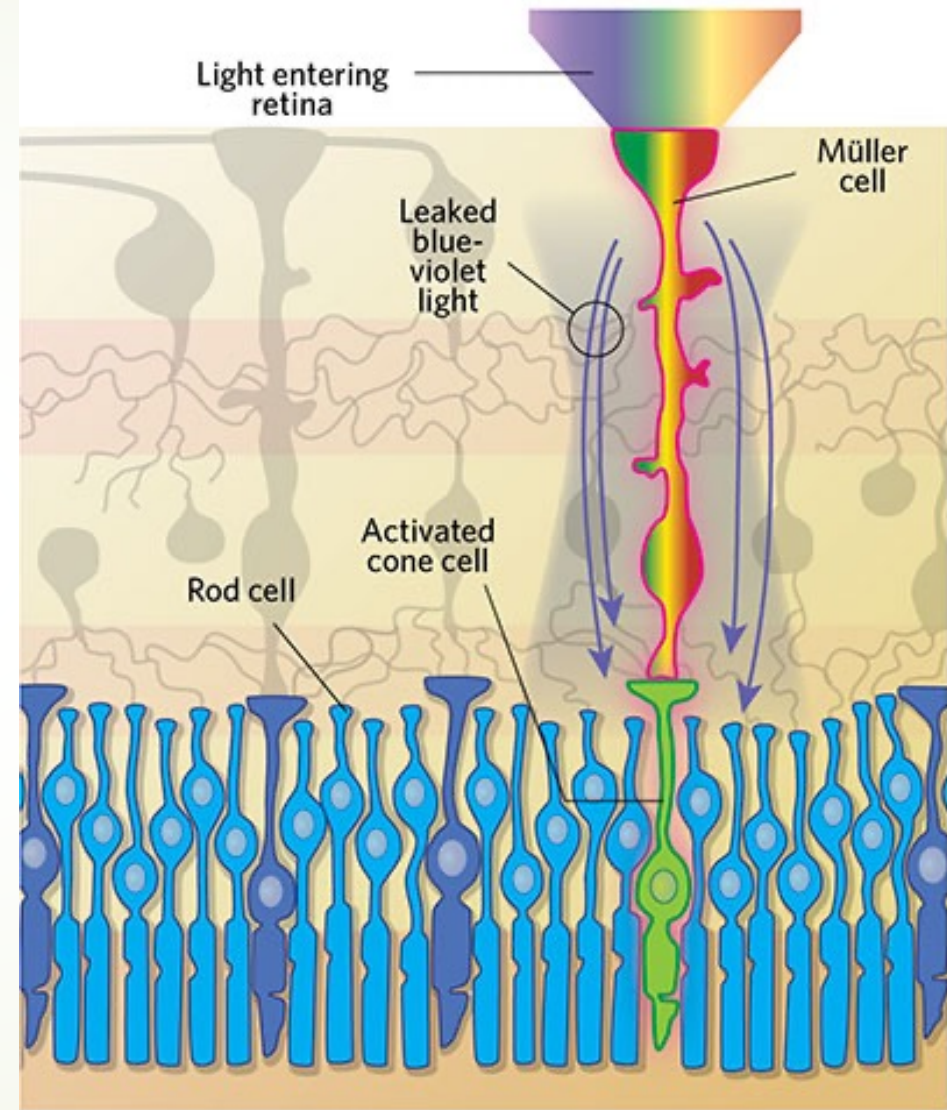
But microglial activation is one of the first events in an array of neurodegenerative diseases including glaucoma.

When activated, microglia change their phenotype markedly and become migratory, antigen-presenting cells that induce the expression of new proteins, cytokines, receptors, and mitogens.

Activated microglia have an altered morphology, may undergo mitosis, and can become scavenging, phagocytic cells with cytotoxic and degradative enzymes.

Müller Glia

- Müller glia are the most common type of glial cell found in the retina. They form the internal limiting membrane of the retina and extend to just deeper than the external limiting membrane, not quite reaching the RPE.
- The major role of the Müller cells is to maintain the structural and functional stability of retinal cells. This includes regulation of the extracellular environment via uptake of potentially toxic neurotransmitters, removal of debris, regulation of K^+ levels, storage of glycogen, electrical insulation of receptors and other neurons, and mechanical support of the neural retina.
- There is also evidence that because of their conical funnel-like shape at the retinal surface they can serve as wave guides for entering light.



Schwann Cells

Schwann cells are the principal glia of the peripheral nervous system (PNS). Glial cells function to support neurons in the PNS.

Some Schwann cells form myelin and some do not. Where they form myelin, they do so in the same manner as done by oligodendroglia.

Major difference is that oligos can provide myelin sheaths for several cells, whereas Schwann cells will provide sheaths for only a single cell. The myelin wrappings, like those formed by oligos, create nodes of Ranvier that speed neural transmission.

